

Laptop Inventory Tracker

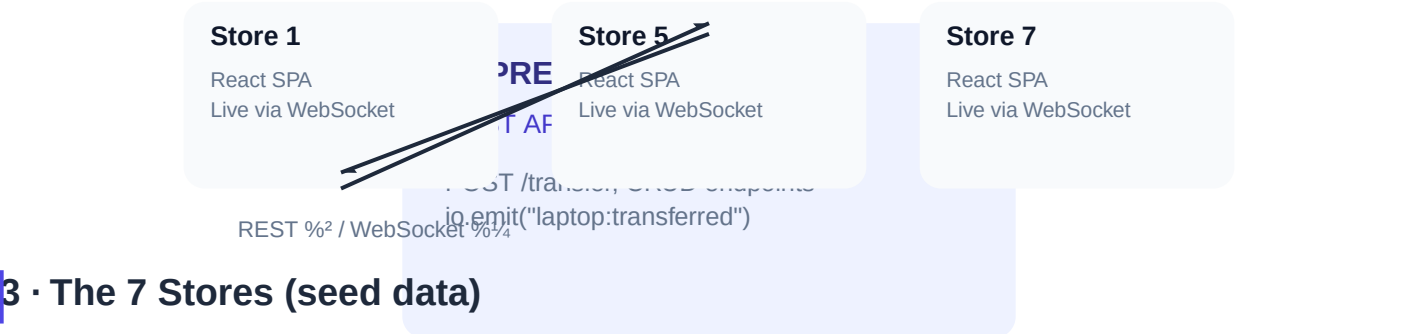
System Map — real-time inventory tracking across 7 retail stores

Node.js · Express · Socket.io · SQLite (node:sqlite) · React 18 · Vite · Tailwind CSS

1 · Overview

A mobile-responsive web app that tracks laptop locations across 7 physical stores. All open screens stay in sync in real time: when an employee changes a laptop's location at Store 1, every other store's screen updates instantly with no refresh. Every transfer is captured in an auditable history log.

2 · Real-Time Architecture



3 · The 7 Stores (seed data)

Store 1: Main Flagship	Store 2: North Hub
Store 3: South Branch	Store 4: East Outlet
Store 5: West Showroom	Store 6: Downtown Express
Store 7: Central Warehouse	

4 · Database Schema (ER)

Primary keys are bold. Foreign keys are drawn with arrows. Transfers are written atomically.

Stores

Column	Type	Notes
id	INTEGER	PK · auto
store_name	TEXT	unique
created_at	TEXT	default now

1 % % N

FK relationships: Laptops.current_store_id != Stores.id · TransferLogs.laptop_id != Laptops.id

Laptops

Column	Type	Notes
id	INTEGER	PK · auto

5 · Real-Time Transfer Flow

What happens when a user changes a laptop's location at any store:

1	laptop_id	TEXT	unique
	current_store_id	INTEGER	FK != Stores.id

User picks a new store in the row dropdown and clicks Confirm Transfer.

2	status	TEXT	In Stock / In Transit / Sold
	created_at	TEXT	default now

React calls POST /api/laptops/:id/transfer { toStoreId }.

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Server runs an atomic SQLite transaction: updates Laptops.current_store_id + inserts a TransferLogs row.

TransferLogs

4			
			Server calls io.emit("laptop:transferred", { laptop, from, to }).
5	laptop_id	INTEGER	FK != Laptops.id
	from_store_id	INTEGER	FK != Stores.id
	to_store_id	INTEGER	FK != Stores.id

Every connected client receives the event and updates that row + history instantly.

6 · Frontend Component Map (React)

App.jsx	state · filters · socket listeners · CRUD handlers
StoreFilter.jsx	filter by store / status · per-store counts
Toolbar.jsx	search box · result count
LaptopTable.jsx	grid · per-row Change Location ! Confirm Transfer · Edit · Delete
HistoryLog.jsx	audit trail of every transfer
InventoryModal.jsx	add / edit a laptop

7 · API & Socket Events

REST Endpoints

GET /api/stores	list the 7 stores
GET /api/laptops?storeId=&status=&search=	filtered inventory
POST /api/laptops	add laptop {brand_model, serial_number, current_store_id, status}
PUT /api/laptops/:id	edit brand / location / status
DELETE /api/laptops/:id	remove laptop
POST /api/laptops/:id/transfer	move laptop {toStoreId}
GET /api/logs	last 100 transfer entries

Socket Events (server !' all clients)

laptop:transferred	{ laptop, from, to } — location change
laptop:created	{ laptop } — new device added
laptop:updated	{ laptop } — device edited
laptop:deleted	{ id } — device removed

Run locally: `cd backend && npm start` · `cd frontend && npm run dev` · <http://localhost:5173>